Aero Vision

SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID** – SIH26074
- **Problem Statement Title** – Downscaling of weather forecast from Block level to Panchayat level
- **Theme** – Agriculture, FoodTech & Rural Development
- **PS Category** – Software
- **Team ID** –
- **Team Name** – Aero-Vision



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AeroAtmos

AI-Powered Downscaling of Weather Forecast from Block Level to Panchayat Level

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CORE CAPABILITIES

01 AI-powered downscaling
Learns each panchayat's correction to the block forecast

02 Multi-source data fusion
Block forecast, 9 km weather history, LGD boundaries, terrain, land cover

03 5-day, IMD-style forecast
Rain, max/min temp, humidity, wind, cloud, chance of rain

04 Tested, not assumed
Scored on panchayats it never saw, against the block forecast

05 Agromet advisories
IMD rainfall categories + heat, frost, humidity & wind rules

06 User-friendly access
FastAPI + web dashboard with real panchayat boundary map

AEROATMOS ECOSYSTEM

INPUT DATA SOURCES

Block forecast
NOAA GFS, 1–5 days ahead (stand-in for IMD feed)

Panchayat boundaries
LGD codes · 87 gram panchayats

Terrain & land cover
Copernicus 90 m DEM · ESA WorldCover 10 m

Weather history
ECMWF IFS 9 km · Jan 2024–Sep 2026

AeroAtmos | AI + ML
Gradient-boosted correction models
Block → Panchayat · 5 variables + chance of rain

PANCHAYAT-LEVEL OUTPUTS · NEXT 5 DAYS

Rainfall (mm) · Max / min temp (°C) · Humidity (%) · Wind speed & direction · Chance of rain · Advisories

Farmers & citizens
Own-panchayat forecast and simple advice

Disaster managers
Heavy-rain, heat & frost alerts; risk map

Agromet units / KVKs
Panchayat inputs for IMD's twice-weekly bulletins

Planned integrations: IMD block feed · SACHET alerts (NDMA) · panchayat rain gauges

VALUE FOR STAKEHOLDERS

Higher accuracy
12–67% lower error than the raw block forecast on unseen panchayats

Early risk awareness
Heavy-rain, heat and frost alerts for each panchayat

Local relevance
Advice for the farmer's own panchayat, not the whole block

Transparent & trusted
Every forecast shows its data source, issue time and tested accuracy

Agriculture benefits
When to irrigate, spray, sow or protect a harvest

Climate resilience
Better-prepared, more resilient villages

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TECHNICAL APPROACH

AeroAtmos — Block-to-Panchayat Weather Downscaling Pipeline

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7-STEP DOWNSCALING PIPELINE

1 — Block forecast input
GFS forecast at each block centre, 1–5 days ahead (IMD feed when available)

2 — Panchayat data
LGD boundaries → location & area; elevation, terrain relief, land cover

3 — Feature building
Own-block forecast + neighbouring blocks (distance-weighted) + terrain + season

4 — ML downscaling
Gradient boosting learns the correction to the block forecast, one model per variable

5 — Validation
Scored on unseen panchayats and the last 6 months against 3 baselines

6 — Advisory generation
IMD rainfall categories + heat, frost, humidity & wind rules; 5-day outlook

7 — API / map output
FastAPI endpoints + dashboard with real panchayat boundary map

SYSTEM ARCHITECTURE

USER / FRONTEND	Web dashboard (Leaflet + Chart.js)	District → Block → Panchayat	5-day forecast & advisory	Boundary map & alerts
AI / ML ENGINE	Python + scikit-learn	Gradient boosting × 5 (corrections)	Chance-of-rain classifier	Validation report
DATA SOURCES	GFS block forecasts (Open-Meteo)	ECMWF IFS 9 km history	LGD panchayat boundaries	Copernicus DEM + ESA WorldCover
STORAGE	CSV / GeoJSON data files	Model file (downscaling.joblib)	Download cache (resumable)	PostgreSQL + PostGIS (planned)
API / OUTPUT	FastAPI (REST)	/forecast — 5-day forecast	/geo — panchayat boundaries	/model/skill · /meta

BUILT FOR SCALE

Hyperlocal
87 real gram panchayats (LGD) today

Data-driven
~2.7 years of real forecasts & weather

Multi-output
5 variables + chance of rain, 5 days ahead

Lightweight
Trains in ~3 minutes on a laptop, no GPU

Scalable
Add any block from LGD with one script

DATA FLOW

Block forecast — GFS today · IMD feed next → **Panchayat features** — LGD · DEM · land cover → **Correction model** — One per variable → **Panchayat forecast** — + IMD-style advisory → **API + dashboard** — Map · alerts · CSV

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FEASIBILITY AND VIABILITY

AeroAtmos – Block-to-Panchayat Weather Downscaling (SIH26074)

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FEASIBLE WITH PROVEN TECHNOLOGY

Proven ML stack
Python, pandas, scikit-learn (gradient boosting), FastAPI, Leaflet

Data available today
Free open data: Open-Meteo (GFS archive, ECMWF IFS, DEM), LGD boundaries, ESA WorldCover

Software-only deployment
No sensors needed; trains in ~3 min on a laptop and runs on any server

KEY CHALLENGES / RISKS

Official IMD data access
IMD's block forecast feed needs approved access; NOAA GFS stands in for now

Ground truth
Checked against ECMWF's 9 km analysis, not rain gauges yet

Rain events
Catches fewer moderate-rain days (32% vs 45%); chance of rain not yet better than the monthly average

Coverage
5 blocks, 87 real gram panchayats in Patna & Gaya

MITIGATION STRATEGIES

Official data integration
Plug in IMD's block feed; the pipeline accepts any block forecast

Ground-truth calibration
Train & validate with IMD AWS and Bihar Mausam Sewa Kendra rain gauges

Rain model upgrade
Separate rain-occurrence model with calibrated probabilities

Expand geography
Build any block's panchayats from LGD boundaries with one script

VALIDATED ON REAL DATA

Truth: ECMWF IFS 9 km analysis · rain days caught 81% vs 64%

Mean absolute error on 22 panchayats never seen in training · test period 26 Mar – 21 Sep 2026

Variable	Our model	Block forecast as-is	Error reduced
Rainfall	**3.8 mm**	4.4 mm	**−12%**
Max temperature	**1.6 °C**	4.4 °C	**−63%**
Min temperature	**1.1 °C**	2.5 °C	**−57%**
Humidity	**6.5 %**	19.6 %	**−67%**
Wind speed	**2.5 km/h**	5.7 km/h	**−56%**

IMPLEMENTATION PATH

✓ Done **Prototype** → ✓ Done **Real data** → ✓ Done **Validation** → Next **Rain-gauge truth** → Next **District-wide**

Safety: prototype forecasts must not be presented as official IMD forecasts.

Reliability: every forecast carries its data source, issue time and tested accuracy.

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IMPACT AND BENEFITS

AeroAtmos – Panchayat-Level Weather Intelligence from Block-Level Forecasts

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WHAT IS AEROATMOS?

An AI/ML system that downscales block-level forecasts to every gram panchayat for the next 5 days, using terrain, land cover and ~2.7 years of real weather data.

Get local forecasts
Panchayat-level, 5 days ahead

View weather parameters
Rain, max/min temp, humidity, wind

See panchayat maps
Real LGD boundaries

Receive actionable advisories
IMD rainfall categories, heat & frost alerts

MULTIDIMENSIONAL IMPACT & BENEFITS

Agriculture & irrigation
When to irrigate, spray or sow; better crop planning; lower input losses

Disaster management
Heavy-rain, heat & frost alerts; panchayat-level readiness

Water management
Local rain estimates for irrigation and reservoir decisions

Community safety
Timely local alerts; better-informed villages

Rural planning
Panchayat-wise weather data for planning and infrastructure

Climate resilience
Adapting farming to local weather risk

KEY BENEFITS

Granular forecasts
Panchayat-level, not block-level

Better planning
Agriculture, water & infrastructure

Farm advisories
Irrigation & spraying

Faster risk awareness
Heavy rain, heat, frost

Scalable
Any block, any district

REAL FORECAST MAP — MASAURHI BLOCK



Sample panchayat forecast — Bara (Masaurhi, Patna)
Wed 30 Sep 2026 · issued 29 Sep, 18:25 IST · real data

0.8 mm	32° / 25°	82 %	6.5 km/h
Very light rain	Max / min °C	Humidity	Wind, from W

Advisory: Dry and calm — a good window for spraying and field work.

Research and References

AeroAtmos – Panchayat-Level Weather Downscaling (SIH26074)

RESEARCH PAPERS

01 **Deep Learning-Based Multi-Hazard Climate Early Warning System Using Machine Learning and Artificial Neural Networks**
Astha Raghuwanshi, Climate Lab, National Institute of Technology Delhi, India

02 **Exploring Large Language Models for Climate Forecasting**
Journal: AIMS Press Atmospheric Science · DOI: 10.3934/aci.202501

DATA SOURCES USED

Open-Meteo API
NOAA GFS forecast archive · ECMWF IFS 9 km · Copernicus DEM — open-meteo.com

LGD boundaries (datameet)
Gram panchayat boundaries & codes — GitHub: ramSeraph/indian_admin_boundaries

ESA WorldCover
10 m land cover — esa-worldcover.org

Planned
IMD MAUSAM API (block feed) · SACHET alerts (NDMA)

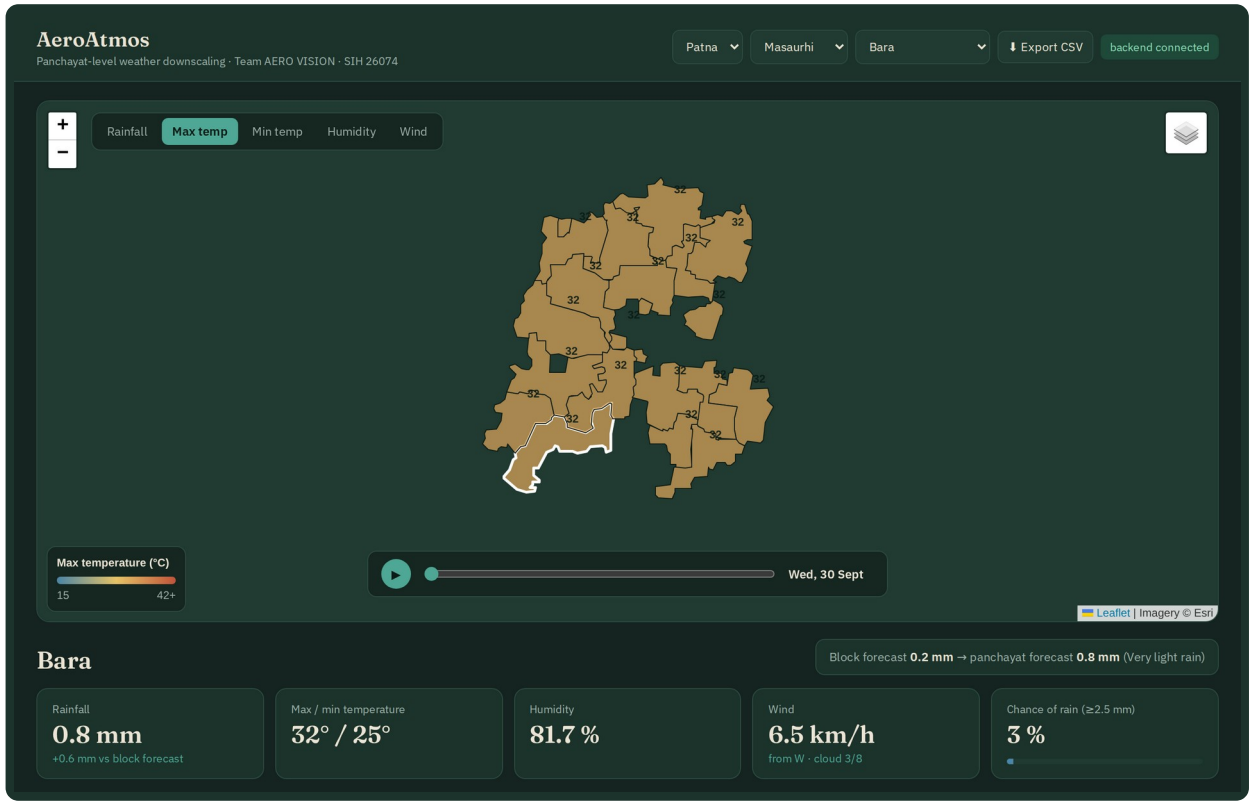
TECHNICAL PLATFORMS

Python + scikit-learn

FastAPI

Leaflet + Chart.js

Planned: Docker, PostgreSQL + PostGIS



AeroAtmos dashboard — real 5-day forecast for Masaurhi block, Patna (issued 29 Sep 2026)

87
real gram panchayats (LGD)

2.7 yrs
of real forecasts & weather

up to 67%
lower error than the block forecast